

HABTAMU MENBERU TEDILA

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EXPERIENCE

- **Postdoctoral researcher**, Radio Astronomy: February 1, 2026 – Present
McGill University, Canada
Supervisor: Prof. Victoria Kaspi
- **Postdoctoral researcher**, Radio Astronomy: May 1, 2023 – January 31, 2026
National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China
Supervisor: Prof. Di Li
- **Lecturer, Physics**: 2014 – 2019
Arba Minch University, Arba Minch, Ethiopia
- **Graduate Assistant II (GA II), Physics**: 2012 – 2014
Arba Minch University, Arba Minch, Ethiopia
- **Graduate Assistant I (GA I), Physics**: 2011 – 2012
Arba Minch University, Arba Minch, Ethiopia

EDUCATION

- **Doctor of Philosophy (Ph.D.), Astrophysics**: 2019 – 2023
University of Chinese Academy of Sciences, Beijing, China
Thesis: Extreme switching and emission variation in radio pulsars
Supervisor: Prof. Na Wang, Prof. Rai Yuen
- **Master of Science (M.Sc.), Astrophysics**: 2012 – 2014
Addis Ababa University, Addis Ababa, Ethiopia
Thesis: Synchrotron radiation from Crab Pulsars
Advisor: Dr. Legesse Wetso Kebede
- **Bachelor of Science (B.Sc.), Physics**: 2008 – 2011
Debre Markos University, Debre Markos, Ethiopia

PUBLICATIONS

2026

1. **Tedila, H. M.**, Li, D., Wang, P., Yuen, R., et al. *Multi-Faceted Emission Properties of PSR J2129+4119 Observed with FAST*, ApJ, 2026, 997(1), 24.

2025

2. **Tedila, H. M.**, Li, D., Wang, P., Yuen, R., Yuan, J. P., Wang, N. et al., *Unveiling the emission properties of three long-period pulsars using FAST*, ApJS, 277(2), 39.
3. Cai, Y., Dang, S. J., **Tedila, H. M.***, Shang, L. H. et al., *the single-pulse study of PSR J0535–0231 discovered by FAST*, MNRAS, 544(1), 299.
4. Yuen, R., **Tedila, H. M.**, *Changes in emission properties associated with the emission shifts in PSR J0344–0901*, Universe, 11(5), 165.
5. Rejep, R., Zhu, W. W., Wang, N. et al. (incl. **Tedila, H. M.**), *A detailed single-pulse study of four-component pulsar PSR J1239+0326 discovered by FAST*, ApJ, 985(1), 11.
6. Khasanov, S., Su, Y., Safarov, A., **Tedila, H. M.**, Mamatkulov, O., *Major radioactive ion beam facilities: A brief global overview*, Eur. Phys. J. Plus, 140, 7.

2024

6. **Tedila, H. M.**, Yuen, R., Wang, N., Li, D. et al., *Investigation of profile shifting and subpulse movement in PSR J0344–0901 with FAST*, ApJ, 965(2), 144.
7. Li, Q., Dang, S., Shang, L., **Tedila, H. M.*** et al., *Subpulse drifting of PSR J1514–4834*, Ap&SS, 369, 90.
8. Dang, S. J., Yuan, J. P., Shang, L. H., Xu, X., **Tedila, H. M.** et al., *Polarization and single-pulse microstructure studies of rotating radio transient J0139+3336 with FAST*, MNRAS, 528(2), 13.
9. Cai, Y. et al. (incl. **Tedila, H. M.**), *The study of mode switching behavior of PSR J0614+2229 using the Parkes ultra-wideband receiver observations*, ApJ, 966(2), 241.

2023

10. Wu, Q. D. et al. (incl. **Tedila, H. M.**), *Follow-up timing of 24 pulsars discovered in the Commensal Radio Astronomy FAST Survey*, MNRAS, 522(4), 5152.
11. Miao, C.-C. et al. (incl. **Tedila, H. M.**), *Reciprocating magnetic fields in the pulsar wind observed from the Black Widow pulsar J1720–0534*, RAA, 23(10), 105005.

2022

12. **Tedila, H. M.**, Yuen, R., Wang, N. et al., *Emission variation of a long-period pulsar discovered by the Five-hundred-meter Aperture Spherical Radio Telescope (FAST)*, ApJ, 929(2), 171.
13. Wen, Z. G. et al. (incl. **Tedila, H. M.**), *A single-pulse study of the subpulse drifter PSR J1631+1252 discovered at FAST*, ApJ, 929(1), 71.

2021

14. **Tedila, H. M.**, Yuen, R., Han, X. H., *Properties of the emission region in pulsars with opposite subpulse drift directions in different profile components*, Ap&SS, 366(10), 100.

15. **Tedila, H. M.**, *Magnetic dipole interaction with multipole magnetic field lines of neutron stars*, Indian J. Phys., 96(3), 689.
16. Khasanov, S., Suleymanov, R., Safarov, A., Safarov, A., **Tedila, H. M.**, Muratov, R., *Calculation of exposure and absorbed dose at irradiation samples by electron bremsstrahlung*, Radiat. Phys. Chem., 188, 109651.

UNDER REVIEW/ACCEPTED PAPERS

1. Dang, S.-J., et al. (including **Tedila, H. M.**) Periodic emission variations of PSR J1857+0943 observed with FAST, The Astrophysical Journal, under review (2026).
2. Cui, X. H., Zhang, C. M., Li, D., Wang et al. (incl. **Tedila, H. M.**), 2025, *Dispersion Measure Evolution in Repeating Fast Radio Bursts with Supernova Remnant-Based Decoupling for Cosmological Applications*, ApJ, Under Review.
3. Zhang, J. S., Wang, T. C. Wang, P., Wu, Q., Li, D. et al. (incl. **Tedila, H. M.**), 2025, *Nature Astronomy*, The magnetar model's energy crisis for a prolific repeating fast radio burst source, Under review.

SUCCESSFUL OBSERVING PROJECT PROPOSALS

2025

1. **Tedila, H. M.**, Li, D., Wang, P. et al., *Follow-up timing studies for four millisecond pulsars discovered by FAST*, PT2025_0207. (PI)

2024

2. **Tedila, H. M.**, Li, D., Wang, P. et al., *Follow-up timing studies for 15 slow-period pulsars discovered by FAST*, PT2024_0220. (PI)
3. **Tedila, H. M.**, Li, D., Wang, P. et al., *Temporal and spectral analysis of microstructures in slow-period CRAFTS pulsars*, PT2024_0036. (PI)
4. Rejep, R. et al. (incl. **Tedila, H. M.**), *Single-pulse study of newly discovered CRAFTS pulsars PSRs J1813–0452 and J0344–0901*, PT2024_0196. (Co-PI)
5. Rejep, R. et al. (incl. **Tedila, H. M.**), *Long-term timing of 6 CRAFTS pulsars discovered by FAST*, PT2024_0248. (Co-PI)

2023

6. **Tedila, H. M.**, Li, D., Wang, P. et al., *Investigation of bi-drifting subpulses in PSR J0535–0231 using FAST telescope*, PT2023_0094. (PI)
7. Rejep, R. et al. (incl. **Tedila, H. M.**), *Single-pulse study of two newly discovered CRAFTS pulsars PSR J2159+0202 and PSR J1239+0326*, PT2023_0179. (Co-PI)
8. Rejep, R. et al. (incl. **Tedila, H. M.**), *Long-term timing of 9 CRAFTS pulsars discovered by FAST*, PT2023_0181. (Co-PI)
9. Dang, S. J., Shang, L. H., Yuan, J. P., **Tedila, H. M.** et al., *Polarization and single-pulse microstructure studies of rotating radio transients with FAST*, PT2023_0129. (Co-PI)

10. Miao, C.-C. et al. (*incl. Tedila, H. M.*), *Timing of 6 millisecond pulsars discovered by CRAFTS*, PT2023_0034. (Co-PI)
11. Li, D. et al. (*incl. Tedila, H. M.*), *FAST newly discovered pulsar follow-up timing observation at Parkes telescope, PX500*. (Co-PI)

2022

12. **Tedila, H. M.**, Yuen, R., Wang, N. et al., *Investigation of the physics of pulsars with complex emission features*, PT2022_0193. (PI)

2020

13. **Tedila, H. M.**, Yuen, R., Wang, N. et al., *Investigating unusual profile mode changing in new FAST-discovered radio pulsars*, PT2020_0052. (PI)
14. Yuan, J. P. et al. (*incl. Tedila, H. M.*), *Timing analysis and emission variations in new pulsars discovered in GPPS*, PT2020_0071. (Co-PI)
15. Yuen, R., Manchester, R. N., Hobbs, G., **Tedila, H. M.** et al., *Extreme switching and radio emission variation in FAST pulsars*, PT2020_0045. (Co-PI)
16. Dang, S. J., Shang, L. H., Yuan, J. P., **Tedila, H. M.** et al., *FAST observation of four “Swooshes” pulsars*, PT2020_0062. (Co-PI)
17. Wang, N. et al. (*incl. Tedila, H. M.*), *Pulsar timing: Physics and evolution of pulsars*, ZD2020_6. (Co-PI)

SELECTED ACADEMIC PRESENTATIONS

2025

1. **The 11th Annual Science at Low Frequencies (SALF) Conference**, oral presentation, Orléans, France, 2025.
2. **The Second FAST Scientific Forum**, oral presentation, Guiyang, China, 2025.

2024

3. **International Pulsar Symposium**, oral presentation, Guiyang, China, 2024.
4. **SKA Pulsar Science Symposium (SPSS)**, oral presentation, Kunming, China, 2024.
5. **Observational Radio Astronomy and Its Future Development in Ethiopia**, invited talk, Addis Ababa, Ethiopia, 2024.
6. **Green Bank Observatory (GBO) Single Dish Summer School**, poster presentation, West Virginia, USA, 2024.

2023

7. **Scientific Program of FAST / Future Pulsar Symposium 12**, oral presentation, Nanyang, China, 2023.

8. **ISM Science Forum**, oral presentation, Wuhan, China, 2023.

2022

9. **FAST Pulsar Symposium 11**, oral presentation, Xiangtan, China, 2022.

OBSERVING EXPERIENCE

- Parkes 64-m Radio telescopes: 2023 – 2026
Observing Expert for the FAST Pulsar Timing follow-up PX500 project
- Nanshan 26-m Radio telescopes: 2019 – 2022
Pulsar observation and monitoring

SELECTED FELLOWSHIPS AND AWARDS

1. 2024 Early-Career Scientist ANSO Visiting Fellowship on Mega-science Facilities
2. 2022 Excellent International Student of the University of Chinese Academy of Sciences (UCAS)
3. 2019 CAS-TWAS President's Fellowship Programme.

MEMBERSHIPS

1. Astronomical Society of Ethiopia (ASE) - Full membership (2025 – present)
2. African Astronomical Society (AfAS) – Full membership (2024 – present)
3. Ethiopian Physical Society (EPS) – Regular membership (2014 – present)

REFERENCES

1. Victoria Kaspi (Professor), Director of McGill University, the Trottier Space Institute 3600 University Street Montreal QC H3A 2T8 Canada
2. Di Li (Professor), Chief Scientist, National Astronomical Observatories, CAS, 20A Datun Road, Chaoyang District, Beijing, 100101, dili@nao.cas.cn
3. Pei Wang (Professor), National Astronomical Observatories, Chinese Academy of Sciences, Postdoctoral research supervisor, A20 Datun Road, Chaoyang District, Beijing 100101, People's Republic of China, email: wangpei@nao.cas.cn
4. Na Wang (Professor), Director, Xinjiang Astronomical Observatory, CAS, 150 Science 1-Street, Urumqi, Xinjiang 830011, People's Republic of China, email: na.wang@xao.ac.cn
5. Rai Yuen (Professor), Researcher and Ph.D supervisor, Xinjiang Astronomical Observatory, CAS, 150 Science 1-Street, Urumqi, Xinjiang 830011, People's Republic of China